

A Generating-Function Framework for the HK-Property in Perfect m -ary Trees

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Abstract

We provide a generating function framework for calculating the sizes of stars in perfect m -ary trees. In particular, we provide an alternate proof of the HK-property for perfect m -ary trees, originally proved by Ākovič and Yan [6]. This framework also quantifies the gap between the leaf star and the root star, giving a structured decomposition of the difference between the two stars. The framework also recovers several EKR consequences for perfect m -ary trees, including formulas for the independence number and for the minimum size of a maximal independent set.

1 Introduction

The Erdős–Ko–Rado theorem [2] is a foundational result about intersecting families of uniform subsets. One graph-theoretic version studies intersecting families of independent sets: for a graph G , one asks whether the largest intersecting subfamilies of independent k -sets are obtained by fixing a vertex. Such fixed-vertex families are called stars. A graph satisfying this extremal property is called k -EKR.

Hurlbert and Kamat [5] proposed a stronger star-centre phenomenon for trees: in any tree, the largest stars should be centred at leaves. This statement, often called the HK-property, is false for general trees, but it remains true for several natural classes. In particular, Ākovič and Yan [6] recently proved the HK-property for all perfect m -ary trees by a combinatorial injection.

This paper develops a complementary approach. Instead of constructing injections between individual star families, we encode the entire distribution of independent sets by generating functions. For the perfect m -ary tree $T_{m,n}$ of depth n , we introduce three polynomials:

$$F_n(x), \quad L_n(x), \quad R_n(x),$$

counting all independent sets, independent sets containing a fixed leaf, and independent sets containing the root, respectively. These polynomials satisfy a coupled recursive system. The HK comparison between leaves and the root then becomes the coefficientwise inequality

$$L_n(x) \geq_c R_n(x).$$

The main contribution is a coefficientwise proof of this inequality for all $m \geq 2$ and all depths n . The proof proceeds by a simultaneous induction driven by two alternating defect sequences. This gives a generating-function proof of the HK-property for perfect m -ary trees.

The same framework also quantifies the gap. Writing

$$D_n(x) := L_n(x) - R_n(x),$$

we derive a recursive decomposition for D_n with an explicitly controlled correction term. Thus the HK-property is not only recovered as a non-negativity statement; the difference between the leaf star and root star is itself shown to satisfy a structured recurrence.

Finally, we record several EKR consequences and related phenomena for perfect m -ary trees. Some of these follow naturally from the structure of the tree, such as the formulas for the independence number and for the minimum size of a maximal independent set. Other results, including low-uniformity classifications and top-layer failures, are logically independent of the generating-function proof and are collected separately.

2 Preliminaries and Notation

Throughout the paper, $m \geq 2$ is fixed. Let $T_{m,n}$ denote the perfect m -ary tree of depth n : the root has depth 0, every non-leaf vertex has exactly m children, and all leaves have depth n . Thus

$$|V(T_{m,n})| = 1 + m + \cdots + m^n = \frac{m^{n+1} - 1}{m - 1}.$$

We use r for the root of $T_{m,n}$ and ℓ for a fixed leaf. By symmetry, the size of a leaf star does not depend on the chosen leaf.

For a graph G , let $V(G)$ and $E(G)$ denote its vertex and edge sets. An *independent set* is a subset of vertices containing no adjacent pair. We define:

$$\mathcal{I}^k(G) = \{I \subseteq V(G) : I \text{ is independent and } |I| = k\}.$$

For $v \in V(G)$, the k -star centred at v is

$$\mathcal{I}_v^k(G) = \{I \in \mathcal{I}^k(G) : v \in I\}.$$

The independence number of G is denoted by $\alpha(G)$.

A family $\mathcal{F} \subseteq \mathcal{I}^k(G)$ is *intersecting* if $A \cap B \neq \emptyset$ for all $A, B \in \mathcal{F}$. The graph G is called *k -EKR* if every intersecting family $\mathcal{F} \subseteq \mathcal{I}^k(G)$ satisfies

$$|\mathcal{F}| \leq \max_{v \in V(G)} |\mathcal{I}_v^k(G)|.$$

It is called *strictly k -EKR* if every maximum-size intersecting family is a star.

We use the coefficientwise partial order on $\mathbb{Z}[x]$. For polynomials $P, Q \in \mathbb{Z}[x]$, we write:

$$P \geq_c Q \iff P - Q \in \mathbb{N}[x].$$

Equivalently, every coefficient of $P - Q$ is non-negative. This order is preserved under addition and under multiplication by polynomials in $\mathbb{N}[x]$. We shall repeatedly use the elementary fact that if $U, V \in \mathbb{N}[x]$ and $U \geq_c V$, then

$$U^a \geq_c V^a \quad (a \geq 1),$$

because

$$U^a - V^a = (U - V) \sum_{j=0}^{a-1} U^{a-1-j} V^j.$$

The HK-property for perfect m -ary trees may be stated as follows.

Definition 2.1 (HK-property). A tree T has the *HK-property* if, for every $k \geq 1$, some leaf ℓ of T satisfies

$$|\mathcal{I}_v^k(T)| \leq |\mathcal{I}_\ell^k(T)| \quad \text{for every } v \in V(T).$$

Because $T_{m,n}$ is symmetric, it is enough to compare a leaf with the root in every rooted perfect subtree.

Lemma 2.2 (Leaf-root domination implies the HK-property). *Assume that for every $a \geq 0$ and every $k \geq 1$, the perfect m -ary tree $T_{m,a}$ satisfies*

$$|\mathcal{I}_r^k(T_{m,a})| \leq |\mathcal{I}_\ell^k(T_{m,a})|,$$

where r is its root and ℓ is any leaf. Then every perfect m -ary tree has the HK-property.

Proof. Let $T = T_{m,n}$ and let $v \in V(T)$. If v is a leaf there is nothing to prove. So, let T_v be the perfect m -ary subtree rooted at v , and let ℓ be a leaf below v . We compare independent k -sets of T containing v with independent k -sets containing ℓ .

For $I \in \mathcal{I}_v^k(T)$, decompose

$$I_1 = I \cap V(T_v), \quad I_0 = I \setminus V(T_v).$$

Fix integers $k_0 + k_1 = k$ and restrict to the sets with $|I_0| = k_0$ and $|I_1| = k_1$. By the hypothesis applied inside T_v , there is an injection from independent k_1 -sets of T_v containing its root v to independent k_1 -sets of T_v containing the leaf ℓ . Applying this injection to I_1 and leaving I_0 unchanged gives an independent k -set of T containing ℓ . The only edge between T_v and the rest of T is the edge from v to its parent, and I_0 is already compatible with I_1 , so the replacement creates no new conflict. Summing over all k_0, k_1 gives

$$|\mathcal{I}_v^k(T)| \leq |\mathcal{I}_\ell^k(T)|.$$

Since all leaves of $T_{m,n}$ are equivalent under automorphisms, the chosen leaf may be replaced by any leaf. $\square$

3 Generating functions

We now introduce the generating functions used throughout the paper. To make the recurrences uniform, we use the convention that $T_{m,-1}$ is the empty tree. Thus

$$F_{-1}(x) = 1, \quad L_{-1}(x) = 0.$$

For $n \geq 0$, define

$$F_n(x) = \sum_{k \geq 0} |\mathcal{I}^k(T_{m,n})| x^k,$$

$$L_n(x) = \sum_{k \geq 0} |\mathcal{I}_\ell^k(T_{m,n})| x^k,$$

and

$$R_n(x) = \sum_{k \geq 0} |\mathcal{I}_r^k(T_{m,n})| x^k.$$

The base values are

$$F_0(x) = 1 + x, \quad L_0(x) = R_0(x) = x.$$

Proposition 3.1 (Generating-function recurrences). *For every $n \geq 1$,*

$$F_n = F_{n-1}^m + xF_{n-2}^{m^2}, \quad (1)$$

$$R_n = xF_{n-2}^{m^2}, \quad (2)$$

$$L_n = L_{n-1}F_{n-1}^{m-1} + xL_{n-2}F_{n-2}^{m^2-1}. \quad (3)$$

Proof. For F_n , condition on whether the root is excluded or included. If the root is excluded, the m child subtrees are independent copies of $T_{m,n-1}$, contributing F_{n-1}^m . If the root is included, its m children are excluded, and the remaining descendants split into m^2 copies of $T_{m,n-2}$. The root contributes a factor of x , giving $xF_{n-2}^{m^2}$.

The formula for R_n is exactly the root-included case.

For L_n , fix a leaf ℓ . If the root is excluded, the child subtree containing ℓ contributes L_{n-1} and the other $m-1$ child subtrees contribute F_{n-1}^{m-1} . If the root is included, all children are excluded; among the m^2 grandchild subtrees, one contains ℓ and contributes L_{n-2} , while the remaining m^2-1 contribute $F_{n-2}^{m^2-1}$. The root again contributes x . $\square$

The coefficientwise HK comparison between a leaf and the root is therefore equivalent to

$$L_n \geq_c R_n.$$

We write

$$D_n := L_n - R_n \quad (n \geq 0)$$

for the HK gap polynomial. Notice that $D_0 = 0$.

4 A coefficientwise proof of the HK-property

The proof uses two alternating coefficientwise defects. These defects are the mechanism that makes the simultaneous induction work.

4.1 Two alternating defects

Lemma 4.1 (First alternating defect). *For $n \geq 2$, define*

$$A_n := F_{n-3}^{m(m-1)} F_{n-1}^{m-1} - F_{n-2}^{m^2-1}.$$

Then

$$A_n \geq_c 0 \quad \text{for odd } n \geq 3,$$

and

$$-A_n \geq_c 0 \quad \text{for even } n \geq 2.$$

Proof. Define

$$C_n := F_{n-3}^m F_{n-1} - F_{n-2}^{m+1} \quad (n \geq 2).$$

Then

$$\begin{aligned} A_n &= (F_{n-3}^m F_{n-1})^{m-1} - (F_{n-2}^{m+1})^{m-1} \\ &= C_n \sum_{j=0}^{m-2} (F_{n-3}^m F_{n-1})^{m-2-j} (F_{n-2}^{m+1})^j. \end{aligned}$$

Thus A_n has the same coefficientwise sign as C_n .

Using (1), for $n \geq 2$ we have

$$\begin{aligned}
C_{n+1} &= F_{n-2}^m F_n - F_{n-1}^{m+1} \\
&= F_{n-2}^m (F_{n-1}^m + x F_{n-2}^{m^2}) - F_{n-1}^{m+1} \\
&= x F_{n-2}^{m(m+1)} - x F_{n-3}^{m^2} F_{n-1}^m \\
&= -x [(F_{n-3}^m F_{n-1})^m - (F_{n-2}^{m+1})^m] \\
&= -x C_n \sum_{j=0}^{m-1} (F_{n-3}^m F_{n-1})^{m-1-j} (F_{n-2}^{m+1})^j.
\end{aligned}$$

Moreover,

$$C_2 = F_{-1}^m F_1 - F_0^{m+1} = F_1 - F_0^{m+1} = -x(F_0^m - 1) \leq_c 0.$$

Hence the coefficientwise sign of C_n alternates with n . Therefore $C_n \geq_c 0$ for odd n and $-C_n \geq_c 0$ for even n . The same sign pattern holds for A_n . $\square$

Lemma 4.2 (Second alternating defect). *For $k \geq 0$, define*

$$J_k := F_k F_{k+1}^m - F_{k-1}^m F_{k+2}.$$

Then

$$J_k \geq_c 0 \quad \text{for even } k,$$

and

$$-J_k \geq_c 0 \quad \text{for odd } k.$$

Proof. For $k \geq 0$, using (1),

$$\begin{aligned}
J_{k+1} &= F_{k+1} F_{k+2}^m - F_k^m F_{k+3} \\
&= F_{k+1} F_{k+2}^m - F_k^m (F_{k+2}^m + x F_{k+1}^{m^2}) \\
&= x F_{k-1}^{m^2} F_{k+2}^m - x F_k^m F_{k+1}^{m^2} \\
&= -x [(F_k F_{k+1})^m - (F_{k-1}^m F_{k+2})^m] \\
&= -x J_k \sum_{j=0}^{m-1} (F_k F_{k+1})^{m-1-j} (F_{k-1}^m F_{k+2})^j.
\end{aligned}$$

The base case is

$$\begin{aligned}
J_0 &= F_0 F_1^m - F_{-1}^m F_2 \\
&= F_0 F_1^m - (F_1^m + x F_0^{m^2}) \\
&= x(F_1^m - F_0^{m^2}) \geq_c 0,
\end{aligned}$$

because $F_1 = F_0^m + x \geq_c F_0^m$. The recurrence for J_{k+1} flips the sign at every step, proving the claim. $\square$

4.2 The main defect recurrence

Lemma 4.3 (Defect recurrence). *For every $n \geq 3$,*

$$D_n = F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} + x F_{n-3}^m A_n. \quad (4)$$

Proof. Using (3), (2), and the definition of D_n ,

$$\begin{aligned} D_n &= L_n - x F_{n-2}^{m^2} \\ &= L_{n-1} F_{n-1}^{m-1} + x L_{n-2} F_{n-2}^{m^2-1} - x F_{n-2}^{m^2}. \end{aligned}$$

For $n \geq 3$,

$$L_{n-1} = D_{n-1} + x F_{n-3}^{m^2}, \quad L_{n-2} = D_{n-2} + x F_{n-4}^{m^2}.$$

Substituting gives

$$\begin{aligned} D_n &= F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} \\ &\quad + x \left[F_{n-3}^{m^2} F_{n-1}^{m-1} + x F_{n-4}^{m^2} F_{n-2}^{m^2-1} - F_{n-2}^{m^2} \right]. \end{aligned}$$

Since $F_{n-2} = F_{n-3}^m + x F_{n-4}^{m^2}$, the bracket simplifies to

$$F_{n-3}^{m^2} F_{n-1}^{m-1} - F_{n-3}^m F_{n-2}^{m^2-1} = F_{n-3}^m A_n.$$

This proves (4). □

We also need two comparison inequalities.

Lemma 4.4 (Even-depth comparison). *For every even $n \geq 4$,*

$$F_{n-1}^{m-1} A_{n-1} + F_{n-3}^{m-1} A_n \geq_c 0.$$

Proof. Expanding and cancelling gives

$$\begin{aligned} &F_{n-1}^{m-1} A_{n-1} + F_{n-3}^{m-1} A_n \\ &= F_{n-2}^{m-1} \left[(F_{n-4}^m F_{n-1})^{m-1} - (F_{n-3} F_{n-2}^m)^{m-1} \right]. \end{aligned}$$

Since n is even, $n-3$ is odd. By Theorem 4.2,

$$J_{n-3} = F_{n-3} F_{n-2}^m - F_{n-4}^m F_{n-1} \leq_c 0.$$

Hence $F_{n-4}^m F_{n-1} \geq_c F_{n-3} F_{n-2}^m$. Taking $(m-1)$ st powers and multiplying by F_{n-2}^{m-1} proves the result. □

Lemma 4.5 (Odd-depth comparison). *For every odd $n \geq 5$,*

$$F_{n-2}^{m^2-1} A_{n-2} - F_{n-4}^{m^2-1} A_n \geq_c 0.$$

Proof. Expanding and cancelling gives

$$\begin{aligned} &F_{n-2}^{m^2-1} A_{n-2} - F_{n-4}^{m^2-1} A_n \\ &= (F_{n-5}^m F_{n-3} F_{n-2}^{m+1})^{m-1} - (F_{n-3}^m F_{n-1} F_{n-4}^{m+1})^{m-1}. \end{aligned}$$

It is enough to compare the two bases. Factor them as

$$F_{n-5}^m F_{n-3} F_{n-2}^{m+1} = (F_{n-5}^m F_{n-2})(F_{n-3} F_{n-2}^m),$$

and

$$F_{n-3}^m F_{n-1} F_{n-4}^{m+1} = (F_{n-4}^m F_{n-3})(F_{n-4}^m F_{n-1}).$$

Since n is odd, $n - 4$ is odd and $n - 3$ is even. By [Theorem 4.2](#),

$$F_{n-5}^m F_{n-2} \geq_c F_{n-4} F_{n-3}^m$$

and

$$F_{n-3} F_{n-2}^m \geq_c F_{n-4}^m F_{n-1}.$$

Multiplying these two inequalities compares the bases. Taking $(m-1)$ st powers gives the lemma. $\square$

4.3 The simultaneous induction

Theorem 4.6 (Coefficientwise HK theorem). *For every $m \geq 2$ and every $n \geq 0$,*

$$L_n(x) \geq_c R_n(x).$$

Equivalently, $D_n(x) \in \mathbb{N}[x]$ for all $n \geq 0$.

Proof. The case $n = 0$ is equality. For $n \geq 1$, we prove simultaneously that

$$D_n \geq_c 0, \tag{5}$$

and the stronger odd-depth assertion

$$D_n \geq_c x F_{n-2} A_n \quad (n \geq 3 \text{ odd}). \tag{6}$$

The ordinary base cases are

$$D_1 = L_1 - R_1 = x(F_0^{m-1} - 1) \geq_c 0,$$

and

$$\begin{aligned} D_2 &= x F_0^{m-1} F_1^{m-1} + x^2 F_0^{m^2-1} - x F_0^m \\ &= x F_0^{m-1} (F_1^{m-1} - F_0^{m(m-1)}) \geq_c 0, \end{aligned}$$

because $F_1 = F_0^m + x \geq_c F_0^m$.

We also need the strong base case $n = 3$. By [Theorem 4.3](#),

$$D_3 = F_2^{m-1} D_2 + x F_1^{m^2-1} D_1 + x F_0^m A_3.$$

Therefore

$$\begin{aligned} D_3 - x F_1 A_3 &= F_2^{m-1} D_2 + x F_1^{m^2-1} D_1 + x (F_0^m - F_1) A_3 \\ &= F_2^{m-1} D_2 + x^2 \left[(F_0^{m-1} - 1) F_1^{m^2-1} - A_3 \right]. \end{aligned}$$

The first term is non-negative. For the bracket, using the definition of A_3 ,

$$\begin{aligned} (F_0^{m-1} - 1) F_1^{m^2-1} - A_3 &= F_0^{m-1} F_1^{m^2-1} - F_0^{m(m-1)} F_2^{m-1} \\ &= (F_0 F_1^{m+1})^{m-1} - (F_0^m F_2)^{m-1}. \end{aligned}$$

It remains to show $F_0 F_1^{m+1} \geq_c F_0^m F_2$, equivalently $F_1^{m+1} \geq_c F_0^{m-1} F_2$. Since $F_2 = F_1^m + x F_0^{m^2}$,

$$\begin{aligned} F_1^{m+1} - F_0^{m-1} F_2 &= F_1^m (F_1 - F_0^{m-1}) - x F_0^{m^2+m-1} \\ &= x \left[F_1^m (F_0^{m-1} + 1) - F_0^{m^2+m-1} \right] \\ &\geq_c x \left[F_0^{m^2} (F_0^{m-1} + 1) - F_0^{m^2+m-1} \right] \\ &= x F_0^{m^2} \geq_c 0. \end{aligned}$$

Thus $D_3 \geq_c x F_1 A_3$, establishing the strong base case.

Now let $n \geq 4$ and assume the relevant assertions for smaller depths. First suppose that n is even. Then $n-1$ is odd, so the strong induction hypothesis gives

$$D_{n-1} \geq_c x F_{n-3} A_{n-1},$$

and the ordinary induction hypothesis gives $D_{n-2} \geq_c 0$. By [Theorem 4.3](#),

$$\begin{aligned} D_n &= F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} + x F_{n-3}^m A_n \\ &\geq_c x F_{n-3} F_{n-1}^{m-1} A_{n-1} + x F_{n-3}^m A_n \\ &= x F_{n-3} (F_{n-1}^{m-1} A_{n-1} + F_{n-3}^{m-1} A_n) \geq_c 0, \end{aligned}$$

where the final inequality is [Theorem 4.4](#). This proves the ordinary assertion for even n .

Now suppose that $n \geq 5$ is odd. We prove the stronger assertion. Using [Theorem 4.3](#),

$$\begin{aligned} D_n - x F_{n-2} A_n &= F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} + x (F_{n-3}^m - F_{n-2}) A_n \\ &= F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} - x^2 F_{n-4}^{m^2} A_n. \end{aligned}$$

By ordinary induction, $D_{n-1} \geq_c 0$. Since $n-2$ is odd, the strong induction hypothesis gives

$$D_{n-2} \geq_c x F_{n-4} A_{n-2}.$$

Therefore

$$D_n - x F_{n-2} A_n \geq_c x^2 F_{n-4} (F_{n-2}^{m^2-1} A_{n-2} - F_{n-4}^{m^2-1} A_n) \geq_c 0,$$

where the final inequality is [Theorem 4.5](#). This proves the strong assertion for odd n , and hence also the ordinary assertion. The simultaneous induction is complete. $\square$

Corollary 4.7 (HK-property for perfect m -ary trees). *Every perfect m -ary tree has the HK-property.*

Proof. By [Theorem 4.6](#), every perfect m -ary tree satisfies leaf-root domination at every depth. The result follows from [Theorem 2.2](#). $\square$

5 Quantifying the HK gap

The proof above already contains a recurrence for the gap polynomial $D_n = L_n - R_n$. We now state it as the structural refinement of the HK-property.

Theorem 5.1 (HK gap decomposition). *For every $n \geq 3$,*

$$D_n = F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} + \Gamma_n,$$

where

$$\Gamma_n = x F_{n-3}^m \left(F_{n-3}^{m(m-1)} F_{n-1}^{m-1} - F_{n-2}^{m^2-1} \right).$$

Moreover,

$$\Gamma_n \geq_c 0 \quad \text{for odd } n \geq 3,$$

and

$$\Gamma_n \leq_c 0 \quad \text{for even } n \geq 4.$$

Proof. The recurrence is [Theorem 4.3](#), with $\Gamma_n = x F_{n-3}^m A_n$. Since $x F_{n-3}^m \in \mathbb{N}[x]$, the sign of Γ_n is the sign of A_n . The parity statement follows from [Theorem 4.1](#). $\square$

Remark 5.2. The gap decomposition is not logically needed after [Theorem 4.6](#); it is a refinement of it. The theorem says that the difference between the leaf star and root star is governed by a second-order inhomogeneous recurrence. The forcing term Γ_n changes sign with the depth, so the non-negativity of D_n is not merely a term-by-term consequence of the decomposition.

For fixed n , write

$$D_n(x) = \sum_{k \geq 0} \delta_{n,k} x^k.$$

Then $\delta_{n,k}$ is the exact difference

$$\delta_{n,k} = |\mathcal{I}_\ell^k(T_{m,n})| - |\mathcal{I}_r^k(T_{m,n})|.$$

Thus [Theorem 5.1](#) gives a recursive way to study the full coefficient sequence of the HK gap.

6 Open problems

The generating-function framework raises several natural questions.

Question 6.1. Is the coefficient sequence of $D_n(x) = L_n(x) - R_n(x)$ unimodal for every $m \geq 2$ and every $n \geq 0$?

Question 6.2. Is $D_n(x)$ log-concave, or real-rooted, for any infinite family of pairs (m, n) ?

Question 6.3. Determine asymptotics for

$$\max_k [x^k] D_n(x)$$

as $n \rightarrow \infty$ with m fixed.

Question 6.4. Give a complete k -EKR classification for perfect m -ary trees. The low-uniformity and upper-layer results above leave a large intermediate range.

Question 6.5. Can the generating-function framework be extended to non-perfect rooted trees, or to broader families of trees for which the HK-property is known or conjectured?

A Proofs for further k -EKR phenomena

Lemma A.1 (Hilton–Milner transfer). *Let G be a graph on N vertices, and let*

$$S_k(G) = \max_{v \in V(G)} |\mathcal{I}_v^k(G)|.$$

If $k \geq 3$ and

$$S_k(G) > \binom{N-1}{k-1} - \binom{N-k-1}{k-1} + 1,$$

then G is strictly k -EKR.

Proof. Let $\mathcal{F} \subseteq \mathcal{I}^k(G)$ be intersecting. If $\bigcap_{F \in \mathcal{F}} F \neq \emptyset$, then $\mathcal{F}$ is contained in a star, so $|\mathcal{F}| \leq S_k(G)$. Otherwise $\mathcal{F}$ is a nontrivial intersecting family of ordinary k -subsets of an N -element set. By the Hilton–Milner theorem [3],

$$|\mathcal{F}| \leq \binom{N-1}{k-1} - \binom{N-k-1}{k-1} + 1 < S_k(G).$$

Thus every maximum intersecting family is a star. $\square$

Proof of Section A. Let ℓ be a leaf of T and let p be its parent. The k -star centred at ℓ is in bijection with independent $(k-1)$ -sets of $T - \{\ell, p\}$. This graph is a forest on $N-2$ vertices with at most $N-3$ edges. Hence

$$S_k(T) \geq \binom{N-2}{k-1} - (N-3) \binom{N-4}{k-3},$$

by subtracting, for each edge, the $(k-1)$ -sets containing that edge. Put $B = \binom{N-2}{k-2}$. Then

$$\binom{N-2}{k-1} = \frac{N-k}{k-1} B$$

and

$$(N-3) \binom{N-4}{k-3} = \frac{(k-2)(N-k)}{N-2} B.$$

Therefore

$$S_k(T) \geq B \cdot \frac{(N-k)(N-2-(k-1)(k-2))}{(k-1)(N-2)}.$$

If $N \geq 2k^2$ and $k \geq 3$, the factor on the right is larger than $k+1$. Indeed, after clearing denominators this is equivalent to positivity of

$$D(N, k) = N^2 - 2Nk^2 + 2Nk - 3N + k^3 - k^2 + 4k - 2.$$

For fixed $k \geq 3$, this polynomial is increasing for $N \geq 2k^2$, and

$$D(2k^2, k) = 5k^3 - 7k^2 + 4k - 2 > 0.$$

Thus $S_k(T) > (k+1)B > kB + 1$. Finally,

$$\binom{N-1}{k-1} - \binom{N-k-1}{k-1} + 1 \leq k \binom{N-2}{k-2} + 1 = kB + 1,$$

by a union bound over the k possible elements of a fixed k -set that another k -set may meet. The result follows from Theorem A.1. The statement for $T_{m,n}$ follows from $N = N_{m,n}$. $\square$

Proof of Section A. A pairwise intersecting family of 2-sets is either contained in a star or consists of the three 2-subsets of a 3-set. For a tree T on N vertices, the independent 2-star centred at v has size $N - 1 - \deg(v)$, so the largest independent 2-star has size $N - 2$ because every nontrivial tree has a leaf.

If $N \geq 6$, then $N - 2 \geq 4$, while every non-star intersecting 2-family has size at most 3; hence T is strictly 2-EKR. If $N = 5$, then the largest star has size 3, and every tree on five vertices has an independent triple, whose three pairs form a non-star intersecting family of size 3. Hence such trees are 2-EKR but not strictly so. If $N = 4$, the only trees are P_4 and $K_{1,3}$. The path P_4 has no independent triple and is strictly 2-EKR, while in $K_{1,3}$ the three leaf-pairs form an intersecting family of size 3, larger than the largest star of size 2. The case $N = 3$ is trivial.

For perfect m -ary trees, $T_{m,1} = K_{1,m}$. Thus the failure case is $(m, n) = (3, 1)$ and the non-strict case is $(m, n) = (4, 1)$. All cases with $n \geq 2$ have at least seven vertices and are strictly 2-EKR. $\square$

Lemma A.2. *Every tree on $N \geq 11$ vertices is strictly 3-EKR.*

Proof. Let ℓ be a leaf with parent p . The 3-star centred at ℓ is in bijection with independent 2-sets of $T - \{\ell, p\}$. This forest has $N - 2$ vertices and at most $N - 3$ edges, so the star has size at least

$$\binom{N-2}{2} - (N-3) = \frac{(N-3)(N-4)}{2}.$$

If $\mathcal{F} \subseteq \mathcal{I}^3(T)$ is intersecting with empty total intersection, the Hilton–Milner theorem [3] gives $|\mathcal{F}| \leq 3N - 8$. For $N \geq 11$,

$$\frac{(N-3)(N-4)}{2} > 3N - 8.$$

Therefore every maximum intersecting family is a star. $\square$

Proof of Section A. If $n \geq 3$, then $|V(T_{m,n})| \geq |V(T_{2,3})| = 15$, so Theorem A.2 gives strict 3-EKR. If $n = 2$, then $|V(T_{m,2})| = 1 + m + m^2$. For $m \geq 3$ this is at least 13, so again the tree is strictly 3-EKR. It remains to handle $T_{2,2}$.

In $T_{2,2}$, let r be the root and let L be the set of four leaves. Then $A = \{r\} \cup L$ is an independent set of size 5, so $\binom{A}{3}$ is an intersecting family of size 10. A leaf-star has size 7: after fixing a leaf, the remaining two vertices must be chosen independently from the five vertices outside that leaf and its parent, and exactly three forbidden pairs remain. The root-star has size $\binom{4}{2} = 6$, and level-one stars are smaller. Hence $T_{2,2}$ is not 3-EKR.

Finally let $n = 1$. Then $T_{m,1} = K_{1,m}$, so independent 3-sets are exactly the 3-subsets of the m leaves. The problem reduces to the ordinary 3-uniform EKR theorem on an m -element ground set. For $m = 3$, there is a unique independent triple, so the tree is strictly 3-EKR. For $m = 4, 5$, the whole layer of triples is intersecting and larger than a star, so the tree is not 3-EKR. For $m = 6$, ordinary EKR gives maximum size $\binom{5}{2} = 10$, and non-star maximum families exist by choosing one triple from each complementary pair. Hence the tree is 3-EKR but not strictly so. For $m \geq 7$, the strict form of the ordinary EKR theorem gives that the only maximum families are stars. $\square$

Lemma A.3 (Large layers are automatically intersecting). *Let G be a graph on N vertices with independence number $\alpha(G)$. If $N/2 < k < \alpha(G)$, then G is not k -EKR.*

Proof. Since $2k > N$, any two k -subsets of $V(G)$ intersect. Hence the whole layer $\mathcal{I}^k(G)$ is an intersecting family. It is not a star: if A is an independent set of size $\alpha(G)$, then for every vertex v there exists an independent k -subset of A avoiding v ; if $v \in A$, use $k < \alpha(G)$. Thus no vertex lies in every member of $\mathcal{I}^k(G)$. $\square$

Lemma A.4 (Expansion from the opposite parity class). *Let $m \geq 3$ and $n \geq 2$. Let*

$$A = \{\text{vertices of } T_{m,n} \text{ at depths } n, n-2, n-4, \dots\}$$

and $B = V(T_{m,n}) \setminus A$. If $\emptyset \neq Y \subseteq B$, then

$$|N(Y)| - |Y| \geq m - 1.$$

Proof. Consider the bipartite forest induced by $Y \cup N(Y)$. Let it have c connected components and e edges. Then

$$e = |Y| + |N(Y)| - c.$$

Every vertex of B has at least m neighbours in A : the root has m children when it lies in B , and every other vertex of B has its parent and its m children in A . Therefore $e \geq m|Y|$. Hence

$$|N(Y)| = e - |Y| + c \geq (m-1)|Y| + c,$$

and so

$$|N(Y)| - |Y| \geq (m-2)|Y| + c \geq m - 1.$$

□

Lemma A.5 (Penultimate expansion). *Let $T_{m,n}$ be a perfect m -ary tree, and let A be the maximum independent set consisting of depths $n, n-2, n-4, \dots$. Put $B = V(T_{m,n}) \setminus A$. If $\emptyset \neq Y \subseteq B$, then*

$$|N(Y)| \geq |Y| + 1.$$

Moreover, equality can occur only when $m = 2$, n is odd, and Y is the singleton consisting of the root.

Proof. Use the same forest-edge count. If $m \geq 3$, every vertex of Y has at least three neighbours in A , so $e \geq 3|Y|$ and hence

$$|N(Y)| = e - |Y| + c \geq 2|Y| + c \geq |Y| + 2.$$

Now suppose $m = 2$. Every non-root vertex of B has three neighbours in A , while the root has two neighbours in A when it belongs to B , which occurs exactly when n is odd. Let $\rho = 1$ if the root belongs to Y , and $\rho = 0$ otherwise. Then $e \geq 3|Y| - \rho$, so

$$|N(Y)| \geq 2|Y| - \rho + c.$$

If $\rho = 0$, this is at least $|Y| + 2$. If $\rho = 1$ and $|Y| = 1$, then Y is the root singleton and equality holds. If $\rho = 1$ and $|Y| \geq 2$, then $|N(Y)| \geq 2|Y| \geq |Y| + 2$. □

Proof of Section A. Part (i) is [Theorem A.3](#) applied to $T_{m,n}$.

For (ii), let A be the maximum independent set consisting of depths $n, n-2, n-4, \dots$, and let $B = V(T_{m,n}) \setminus A$. Let $I \in \mathcal{I}^{\alpha_{m,n}-t}(T_{m,n})$ and set $Y = I \cap B$. Since vertices selected from A must avoid $N(Y)$,

$$|I| \leq |Y| + \alpha_{m,n} - |N(Y)| = \alpha_{m,n} - (|N(Y)| - |Y|).$$

If $Y \neq \emptyset$, [Theorem A.4](#) gives $|I| \leq \alpha_{m,n} - (m-1)$. Thus for $t < m-1$, every independent $(\alpha_{m,n}-t)$ -set is contained in A . At $t = m-1$, every such set has at least $\alpha_{m,n} - (m-1)$ vertices from A . Since $n \geq 2$, $\alpha_{m,n} \geq m^2 + 1$, and therefore $\alpha_{m,n} - t > \alpha_{m,n}/2$ for $1 \leq t \leq m-1$. Hence

the whole layer is intersecting. It is not a star, since for every vertex v there is an independent set in the layer avoiding v : choose from $A \setminus \{v\}$ if $v \in A$, and from A if $v \in B$.

For (iii), use [Theorem A.5](#). If $I \in \mathcal{I}^{\alpha_{m,n}-1}(T_{m,n})$ and $Y = I \cap B$, then

$$|I| \leq |Y| + \alpha_{m,n} - |N(Y)|.$$

Thus every independent $(\alpha_{m,n} - 1)$ -set is either of the form $A \setminus \{a\}$, or, in the exceptional case $m = 2$ and odd n , may also be the set consisting of the root together with $A \setminus N(r)$. These sets are pairwise intersecting when $\alpha_{m,n} \geq 3$. The whole layer is not a star, because each vertex is avoided by some member of the layer. Hence $T_{m,n}$ is not $(\alpha_{m,n} - 1)$ -EKR. $\square$

References

- [1] P. Borg and F. C. Holroyd, The Erdős–Ko–Rado property of various graphs containing singletons, *Discrete Mathematics* 309 (2009), 2877–2885.
- [2] P. Erdős, C. Ko, and R. Rado, Intersection theorems for systems of finite sets, *Quarterly Journal of Mathematics* 12 (1961), 313–320.
- [3] A. J. W. Hilton and E. C. Milner, Some intersection theorems for systems of finite sets, *Quarterly Journal of Mathematics* 18 (1967), 369–384.
- [4] F. C. Holroyd and J. Talbot, Graphs with the Erdős–Ko–Rado property, *Discrete Mathematics* 293 (2005), 165–176.
- [5] G. Hurlbert and V. Kamat, Erdős–Ko–Rado theorems for chordal graphs and trees, *Journal of Combinatorics* 2 (2011), 87–116.
- [6] J. Ľkovič and Y. Yan, The Hurlbert–Kamat property for perfect m -ary trees, preprint, 2026.